

SECTION 1: DOUBLE RIVETED LAP JOINT UNDER TENSION, SHEAR, AND CRUSHING STRESSES

System Parameters: Plate Thickness $t = 15$ mm, Rivet Diameter $d = 25$ mm, Pitch $p = 75$ mm, Ultimate Tensile Stress $\sigma_u = 400$ MPa, Ultimate Shear Stress $\tau_u = 320$ MPa, Ultimate Crushing Stress $\sigma_{cu} = 640$ MPa, Factor of Safety FOS = 4.

Design Protocol: Evaluate plate tearing, rivet shearing ($n = 2$), and rivet crushing strengths to determine minimum rupturing force P_u per pitch; compute working load $P = P_u / \text{FOS}$ and evaluate working stresses.

1. Tearing Strength of Plate (P_t)
Plate tearing capacity per pitch length

$$\begin{aligned} P_t &= (p - d)t \cdot \sigma_u \\ &= (75 - 25) \times 15 \times 400 \\ &= 300000 \text{ N} \\ \Rightarrow P_t &= 300 \text{ kN} \end{aligned}$$

2. Shearing Strength of Rivets (P_s)
Single shear capacity of 2 rivets per pitch length

$$\begin{aligned} P_s &= n \cdot \frac{\pi}{4} d^2 \cdot \tau_u \\ &= 2 \times \frac{\pi}{4} (25)^2 \times 320 \\ &= 314160 \text{ N} \\ \Rightarrow P_s &= 314.16 \text{ kN} \end{aligned}$$

3. Crushing Strength of Rivets (P_c)
Bearing capacity of 2 rivets per pitch length

$$\begin{aligned} P_c &= n \cdot d \cdot t \cdot \sigma_{cu} \\ &= 2 \times 25 \times 15 \times 640 \\ &= 480000 \text{ N} \\ \Rightarrow P_c &= 480 \text{ kN} \end{aligned}$$

4. Rupturing Force per Pitch (P_u)
Minimum strength governing joint rupture

$$\begin{aligned} P_u &= \min(P_t, P_s, P_c) \\ &= \min(300, 314.16, 480) \\ \Rightarrow P_u &= 300 \text{ kN} \end{aligned}$$

5. Safe Working Load (P)
Applied force per pitch at FOS = 4

$$\begin{aligned} P &= \frac{P_u}{\text{FOS}} \\ &= \frac{300}{4} \\ &= 75 \text{ kN} \\ &= 75000 \text{ N} \end{aligned}$$

6. Actual Tensile Stress (σ_t)
Induced tensile stress on main plate

$$\begin{aligned} \sigma_t &= \frac{P}{(p - d)t} \\ &= \frac{75000}{(75 - 25) \times 15} \\ &= 100 \text{ MPa} \end{aligned}$$

7. Actual Shear Stress (τ)

Induced shear stress in rivets

$$\begin{aligned}\tau &= \frac{P}{n \cdot \frac{\pi}{4} d^2} \\ &= \frac{75000}{2 \times \frac{\pi}{4} (25)^2} \\ &= 76.4 \text{ MPa}\end{aligned}$$

8. Actual Crushing Stress (σ_c)

Induced crushing stress on rivets

$$\begin{aligned}\sigma_c &= \frac{P}{n \cdot d \cdot t} \\ &= \frac{75000}{2 \times 25 \times 15} \\ &= 100 \text{ MPa}\end{aligned}$$

Design Output

**Rupturing Force $P_u = 300 \text{ kN}$ | Working Load $P = 75 \text{ kN}$
| Working Stresses: $\sigma_t = 100 \text{ MPa}$, $\tau = 76.4 \text{ MPa}$,
 $\sigma_c = 100 \text{ MPa}$**

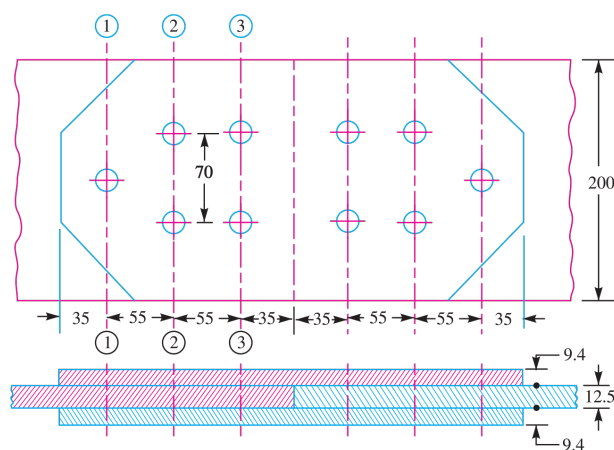


Fig. 9.20. All dimensions in mm.

Fig. 9.20: Double Riveted Lap Joint Scheme

SECTION 2: EFFICIENCY OF SINGLE AND DOUBLE RIVETED LAP JOINTS

System Parameters: Plate Thickness $t = 6$ mm, Rivet Diameter $d = 20$ mm, Case 1 Pitch $p_1 = 50$ mm ($n = 1$), Case 2 Pitch $p_2 = 65$ mm ($n = 2$), Allowable Stresses: Tensile $\sigma_t = 120$ MPa, Shear $\tau = 90$ MPa, Crushing $\sigma_c = 180$ MPa.

Design Protocol: Calculate un-punched solid plate strength, tearing, shearing, and crushing strengths for single and double riveted lap joints; compute joint efficiencies.

1. Solid Plate Strength — Single Lap (P) Strength of un-punched plate ($p = 50$ mm)	$\begin{aligned} P &= p \cdot t \cdot \sigma_t \\ &= 50 \times 6 \times 120 \\ &= 36000 \text{ N} \end{aligned}$
2. Tearing Strength — Single Lap (P_t) Net section tensile capacity	$\begin{aligned} P_t &= (p - d)t \cdot \sigma_t \\ &= (50 - 20) \times 6 \times 120 \\ &= 21600 \text{ N} \end{aligned}$
3. Shearing Strength — Single Lap (P_s) Single shear capacity ($n = 1$)	$\begin{aligned} P_s &= n \cdot \frac{\pi}{4} d^2 \cdot \tau \\ &= 1 \times \frac{\pi}{4} (20)^2 \times 90 \\ &= 28274 \text{ N} \end{aligned}$
4. Crushing Strength — Single Lap (P_c) Bearing capacity ($n = 1$)	$\begin{aligned} P_c &= n \cdot d \cdot t \cdot \sigma_c \\ &= 1 \times 20 \times 6 \times 180 \\ &= 21600 \text{ N} \end{aligned}$
5. Efficiency — Single Riveted Lap Joint (η_1) Ratio of minimum strength to solid strength	$\begin{aligned} \eta_1 &= \frac{\min(P_t, P_s, P_c)}{P} \times 100\% \\ &= \frac{21600}{36000} \times 100\% \\ &\Rightarrow \eta_1 = 60.0\% \end{aligned}$
6. Solid Plate Strength — Double Lap (P) Strength of un-punched plate ($p = 65$ mm)	$\begin{aligned} P &= p \cdot t \cdot \sigma_t \\ &= 65 \times 6 \times 120 \\ &= 46800 \text{ N} \end{aligned}$
7. Tearing Strength — Double Lap (P_t) Net section tensile capacity	$\begin{aligned} P_t &= (p - d)t \cdot \sigma_t \\ &= (65 - 20) \times 6 \times 120 \\ &= 32400 \text{ N} \end{aligned}$

SECTION 3: EFFICIENCY OF DOUBLE RIVETED DOUBLE COVER BUTT JOINT

System Parameters: Plate Thickness $t = 20$ mm, Rivet Diameter $d = 25$ mm, Pitch $p = 100$ mm, Allowable Stresses: Tensile $\sigma_t = 120$ MPa, Shear $\tau = 100$ MPa, Crushing $\sigma_c = 150$ MPa, Double Shear Factor = 2.0.

Design Protocol: Evaluate solid plate strength, plate tearing strength, rivet double shear strength ($n = 2$), and rivet crushing strength to determine joint efficiency.

1. Solid Plate Strength (P) Un-punched solid plate tensile capacity	$\begin{aligned} P &= p \cdot t \cdot \sigma_t \\ &= 100 \times 20 \times 120 \\ &= 240000 \text{ N} \end{aligned}$
2. Tearing Strength of Plate (P_t) Net section tensile strength per pitch	$\begin{aligned} P_t &= (p - d)t \cdot \sigma_t \\ &= (100 - 25) \times 20 \times 120 \\ &= 180000 \text{ N} \end{aligned}$
3. Shearing Strength of Rivets (P_s) Double shear capacity of 2 rivets per pitch ($n = 2$, factor = 2)	$\begin{aligned} P_s &= 2 \cdot \left(2 \cdot \frac{\pi}{4} d^2 \right) \cdot \tau \\ &= 2 \times \left(2 \times \frac{\pi}{4} (25)^2 \right) \times 100 \\ &= 196350 \text{ N} \end{aligned}$
4. Crushing Strength of Rivets (P_c) Bearing capacity of 2 rivets per pitch ($n = 2$)	$\begin{aligned} P_c &= n \cdot d \cdot t \cdot \sigma_c \\ &= 2 \times 25 \times 20 \times 150 \\ &= 150000 \text{ N} \end{aligned}$
5. Joint Efficiency (η) Ratio of minimum strength to un-punched solid plate strength	$\begin{aligned} \eta &= \frac{\min(P_t, P_s, P_c)}{P} \times 100\% \\ &= \frac{150000}{240000} \times 100\% \\ &\Rightarrow \eta = 62.5\% \end{aligned}$
Design Output	Double Cover Butt Joint Efficiency eta = 62.5% (Governed by Rivet Crushing)

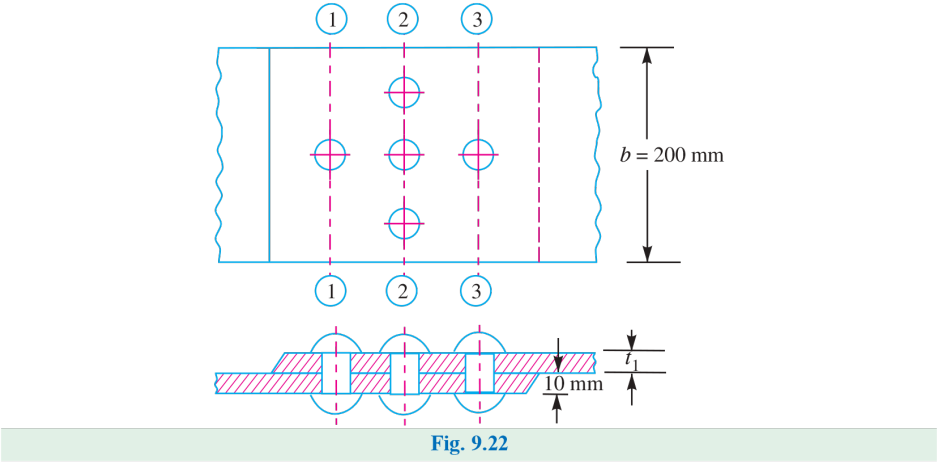


Fig. 9.22: Double Riveted Double Cover Butt Joint

SECTION 4: DOUBLE RIVETED LAP JOINT — ZIG-ZAG RIVETING

System Parameters: Plate Thickness $t = 13\text{ mm}$, Tensile Stress $\sigma_t = 80\text{ MPa}$, Shear Stress $\tau = 60\text{ MPa}$, Crushing Stress $\sigma_c = 120\text{ MPa}$, Rivet Rows $n = 2$ (zig-zag).

Design Protocol: Calculate rivet hole diameter d via Unwin’s formula, determine pitch p by equating $P_t = P_s$ and verify $p \leq p_{\text{max}}$, compute row pitch p_b and margin m , evaluate governing failure mode ($\min(P_t, P_s, P_c)$), and calculate joint efficiency η .

1. Diameter of Rivet Hole (d) Unwin’s formula for $t = 13\text{ mm} > 8\text{ mm}$ and IS standard size	$d = 6\sqrt{t}$ $= 6\sqrt{13} = 21.63\text{ mm}$ $\Rightarrow \text{Standard Hole } d = 23\text{ mm}$ $\Rightarrow \text{Rivet Dia } d_0 = 22\text{ mm}$
2. Pitch of Rivets (p) Equating tearing strength P_t to single shear strength P_s ($n = 2$)	$P_t = (p - d)t \cdot \sigma_t = (p - 23)1040\text{ N}$ $P_s = 2 \times \frac{\pi}{4}(23)^2 \times 60 = 49864\text{ N}$ $(p - 23)1040 = 49864$ $p - 23 = 48 \Rightarrow p = 71\text{ mm}$ $p_{\text{max}} = 2.62t + 41.28 = 75.34\text{ mm}$ $\Rightarrow \text{Adopt Pitch } p = 71\text{ mm}$
3. Distance Between Rows & Margin (p_b, m) Row pitch for zig-zag arrangement and margin	$p_b = 0.33p + 0.67d$ $= 0.33(71) + 0.67(23) = 38.8\text{ mm}$ $\Rightarrow \text{Adopt Row Pitch } p_b = 40\text{ mm}$ $m = 1.5d = 1.5(23) = 34.5\text{ mm}$ $\Rightarrow \text{Adopt Margin } m = 35\text{ mm}$
4. Failure Mode Analysis Tearing (P_t), shearing (P_s), and crushing (P_c) capacities	$P_t = (71 - 23)(13)(80) = 49920\text{ N}$ $P_s = 2\left(\frac{\pi}{4}\right)(23)^2(60) = 49864\text{ N}$ $P_c = 2(23)(13)(120) = 71760\text{ N}$ $\Rightarrow \text{Governing Strength } P_u = P_s = 49864\text{ N}$ $\Rightarrow \text{Failure Mode: Shearing of Rivets}$
5. Joint Efficiency (η) Solid plate strength P and joint efficiency	$P = p \cdot t \cdot \sigma_t = 71(13)(80) = 73840\text{ N}$ $\eta = \frac{P_s}{P} \times 100\%$ $= \frac{49864}{73840} \times 100\%$ $\Rightarrow \eta = 67.5\%$

Rivet Hole $d = 23\text{ mm}$ | Pitch $p = 71\text{ mm}$ | Row Pitch $p_b = 40\text{ mm}$ | Margin $m = 35\text{ mm}$ | Failure: Shearing | Efficiency $\eta = 67.5\%$

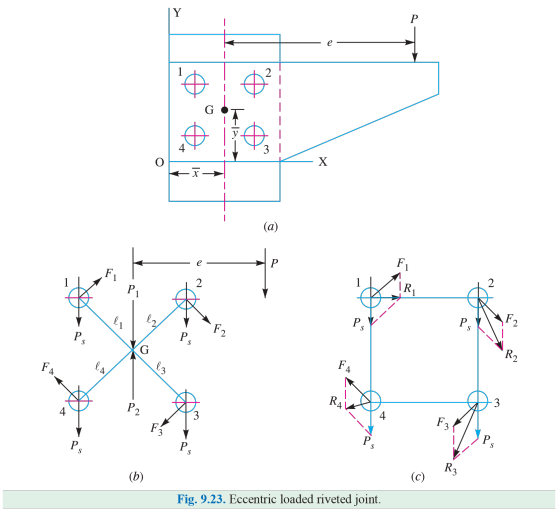


Fig. 9.23: Double Riveted Lap Joint with Zig-Zag Riveting

SECTION 5: TRIPLE RIVETED LAP JOINT — ZIG-ZAG PATTERN

System Parameters: Plate Thickness $t = 7$ mm, Tensile Stress $\sigma_t = 90$ MPa, Shear Stress $\tau = 60$ MPa, Crushing Stress $\sigma_c = 120$ MPa, Rivet Rows $n = 3$ (zig-zag).

Design Protocol: For thin plates ($t < 8$ mm), equate shearing strength to crushing strength ($P_s = P_c$) to determine rivet hole diameter d , equate tearing strength to shearing strength ($P_t = P_s$) to solve for pitch p , limit pitch by I.B.R. maximum pitch ($p \leq p_{\max}$), calculate row pitch p_b , and evaluate governing failure mode.

1. Rivet Diameter (d) Equating shearing strength P_s to crushing strength P_c ($t = 7$ mm $<$ 8 mm)	$141.4d^2 = 2520d$ $d = \frac{2520}{141.4} = 17.8 \text{ mm}$ $\Rightarrow \text{Standard Hole } d = 19 \text{ mm}$ $\Rightarrow \text{Rivet Dia } d_0 = 18 \text{ mm}$
2. Rivet Pitch & Max Pitch ($p, p_{\max}$) Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit	$630(p - 19) = 51045 \text{ N}$ $p - 19 = \frac{51045}{630} = 81 \Rightarrow p = 100 \text{ mm}$ $p_{\max} = 3.47t + 41.28 = 65.57 \text{ mm}$ $\Rightarrow \text{Adopt Pitch } p = p_{\max} = 66 \text{ mm}$
3. Distance Between Rows (p_b) Transverse row pitch for zig-zag riveting	$p_b = 0.33p + 0.67d$ $= 0.33(66) + 0.67(19)$ $= 21.78 + 12.73$ $\Rightarrow \text{Row Pitch } p_b = 34.5 \text{ mm}$
4. Mode of Failure Analysis Tearing (P_t), shearing (P_s), and crushing (P_c) capacities for adopted $p = 66$ mm	$P_t = (66 - 19)(7)(90) = 29610 \text{ N}$ $P_s = 3\left(\frac{\pi}{4}\right)(19)^2(60) = 51045 \text{ N}$ $P_c = 3(19)(7)(120) = 47880 \text{ N}$ $\Rightarrow \text{Governing Strength } P_u = P_t = 29610 \text{ N}$ $\Rightarrow \text{Failure Mode: Tearing off the Plate}$
Design Output Summary	Rivet Hole $d = 19$ mm Adopted Pitch $p = 66$ mm Row Pitch $p_b = 34.5$ mm Failure Mode: Tearing of Plate

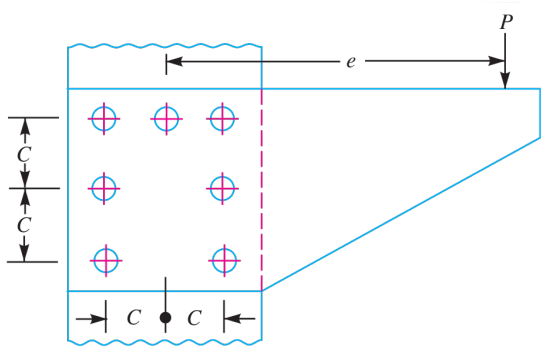


Fig. 9.24

Fig. 9.24: Triple Riveted Lap Joint with Zig-Zag Pattern

SECTION 6: SINGLE RIVETED DOUBLE STRAP BUTT JOINT

System Parameters: Main Plate Thickness $t = 10\text{ mm}$, Tensile Stress $\sigma_t = 80\text{ MPa}$, Shear Stress $\tau = 60\text{ MPa}$, Double Shear Factor = 1.875, Number of Rivets $n = 1$.

Design Protocol: Calculate rivet hole diameter d via Unwin’s formula, determine double shear capacity P_s , solve for pitch p by equating $P_t = P_s$ and verify against I.B.R. maximum pitch ($p \leq p_{\max}$), compute cover strap thickness $t_1 = 0.625t$, evaluate governing failure mode, and calculate efficiency η .

1. Diameter of Rivet Hole (d) Unwin’s formula for $t = 10\text{ mm} > 8\text{ mm}$ and IS standard size	$d = 6\sqrt{t}$ $= 6\sqrt{10} = 18.97\text{ mm}$ $\Rightarrow \text{Standard Hole } d = 19\text{ mm}$ $\Rightarrow \text{Rivet Dia } d_0 = 18\text{ mm}$
2. Double Shear Strength (P_s) Double shear capacity for single rivet ($n = 1$)	$P_s = 1.875 \cdot n \cdot \frac{\pi}{4} d^2 \cdot \tau$ $= 1.875 \times 1 \times \frac{\pi}{4} (19)^2 \times 60$ $= 31900\text{ N}$
3. Pitch & Max Pitch Limit ($p, p_{\max}$) Equating $P_t = P_s$ and checking against I.B.R. maximum pitch	$800(p - 19) = 31900\text{ N}$ $p - 19 = \frac{31900}{800} = 39.87 \Rightarrow p = 58.87\text{ mm}$ $p_{\max} = 1.75t + 41.28 = 58.78\text{ mm}$ $\Rightarrow \text{Adopt Pitch } p = p_{\max} = 60\text{ mm}$
4. Cover Strap Thickness (t_1) Standard proportion for double cover straps	$t_1 = 0.625t$ $= 0.625 \times 10$ $\Rightarrow \text{Strap Thickness } t_1 = 6.25\text{ mm}$
5. Failure Mode & Joint Efficiency (η) Tearing (P_t), shearing (P_s), solid plate strength (P), and efficiency	$P_t = (60 - 19)(10)(80) = 32800\text{ N}$ $P_s = 31900\text{ N} \Rightarrow \text{Failure Mode: Shearing of Rivets}$ $P = 60 \times 10 \times 80 = 48000\text{ N}$ $\eta = \frac{31900}{48000} \times 100\%$ $\Rightarrow \eta = 66.5\%$
Design Output Summary	Rivet Hole $d = 19\text{ mm}$ Pitch $p = 60\text{ mm}$ Cover Strap $t_1 = 6.25\text{ mm}$ Failure Mode: Shearing Efficiency $\eta = 66.5\%$

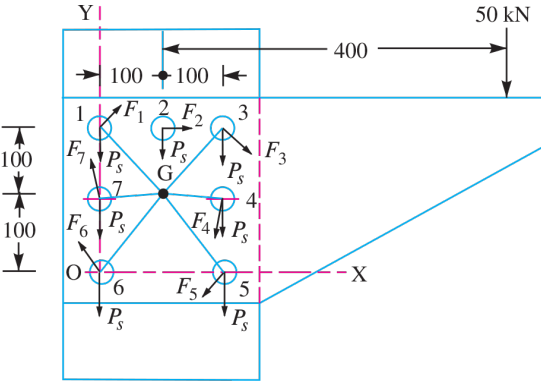


Fig. 9.25

Fig. 9.25: Single Riveted Double Strap Butt Joint

SECTION 7: BOILER LONGITUDINAL DOUBLE RIVETED BUTT JOINT

System Parameters: Boiler Shell Diameter $D = 1.5 \text{ m} = 1500 \text{ mm}$, Steam Pressure $P = 0.95\text{N/mm}^2$, Target Efficiency $\eta_l = 75\%$, Tensile Stress $\sigma_t = 90 \text{ MPa}$, Crushing Stress $\sigma_c = 140 \text{ MPa}$, Shear Stress $\tau = 56 \text{ MPa}$, Double Shear Factor = 1.875, Rivet Rows $n = 2$.

Design Protocol: Calculate shell thickness t , size rivet hole diameter d , determine pitch p by equating $P_t = P_s$ and limit by I.B.R. max pitch ($p \leq p_{\max}$), calculate row pitch p_b , strap thickness t_1 , and margin m , evaluate failure capacities (P_t, P_s, P_c), and verify joint efficiency η .

1. Shell Plate Thickness (t) Thin cylinder formula with 1 mm allowance	$t = \frac{P \cdot D}{2\sigma_t \cdot \eta_l} + 1 \text{ mm}$$= \frac{0.95 \times 1500}{2 \times 90 \times 0.75} + 1$$= \frac{1425}{135} + 1 = 10.6 + 1$$\Rightarrow \text{Shell Thickness } t = 12 \text{ mm}$
2. Diameter of Rivet Hole (d) Unwin's formula for $t = 12 \text{ mm} > 8 \text{ mm}$ and IS standard size	$d = 6\sqrt{t}$$= 6\sqrt{12} = 20.78 \text{ mm}$$\Rightarrow \text{Standard Hole } d = 21 \text{ mm}$$\Rightarrow \text{Rivet Dia } d_0 = 20 \text{ mm}$
3. Pitch & Max Pitch Limit ($p, p_{\max}$) Equating $P_t = P_s$ ($n = 2$ double shear) and checking I.B.R. maximum pitch	$P_s = 2(1.875)\left(\frac{\pi}{4}\right)(21)^2(56) = 72745 \text{ N}$$1080(p - 21) = 72745 \Rightarrow p = 88.35 \text{ mm}$$p_{\max} = 3.5t + 41.28 = 3.5(12) + 41.28 = 83.28 \text{ mm}$$\Rightarrow \text{Adopt Pitch } p = p_{\max} = 84 \text{ mm}$
4. Row Pitch, Strap Thickness & Margin (p_b, t_1, m) Proportions for zig-zag double cover butt joint	$p_b = 0.33p + 0.67d = 0.33(84) + 0.67(21) = 41.8 \text{ mm}$$\Rightarrow \text{Adopt Row Pitch } p_b = 42 \text{ mm}$$t_1 = 0.625t = 0.625(12) \Rightarrow \text{Strap Thickness } t_1 = 7.5 \text{ mm}$$m = 1.5d = 1.5(21) = 31.5 \text{ mm} \Rightarrow \text{Margin } m = 32 \text{ mm}$
5. Failure Mode & Joint Efficiency Verification (η) Capacities for $p = 84 \text{ mm}$, solid plate strength P, and efficiency	$P_t = (84 - 21)(12)(90) = 68040 \text{ N}$$P_s = 72745 \text{ N}, \quad P_c = 2(21)(12)(140) = 70560 \text{ N}$$\Rightarrow \text{Governing Strength } P_u = P_t = 68040 \text{ N (Tearing)}$$P = 84(12)(90) = 90720 \text{ N}$$\eta = \frac{68040}{90720} \times 100\% \Rightarrow \eta = 75.0\%$

Shell t = 12 mm | Hole d = 21 mm | Pitch p = 84 mm |
Row Pitch pb = 42 mm | Strap t1 = 7.5 mm | Margin m =
32 mm | Efficiency eta = 75%

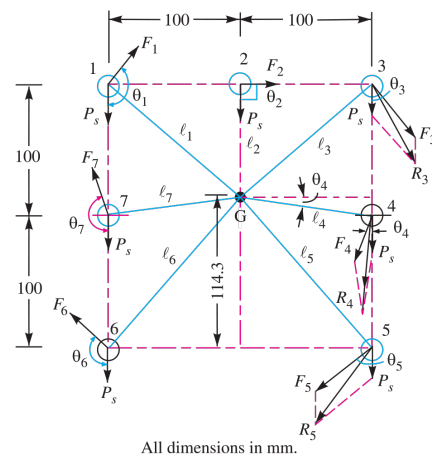


Fig. 9.26: Boiler Longitudinal Double Riveted Double Strap Butt Joint

SECTION 8: PRESSURE VESSEL DOUBLE STRAP BUTT JOINT

System Parameters: Vessel Diameter $D = 1\text{ m} = 1000\text{ mm}$, Pressure $p_i = 2.75\text{N/mm}^2$, Target Efficiency $\eta_l = 79\%$, Allowable Tensile Stress $\sigma_t = 88\text{ MPa}$, Allowable Shear Stress $\tau = 64\text{ MPa}$, Double Shear Factor = 1.8.

Design Protocol: Size shell thickness t , solve rivet diameter d , determine outer and inner row pitches ($p_{\text{outer}}, p_{\text{inner}}$), and verify efficiency.

1. Shell Plate Thickness (t) Thin cylinder formula	$t = \frac{p_i \cdot D}{2\sigma_t \cdot \eta_l}$$= \frac{2.75 \times 1000}{2 \times 88 \times 0.79}$$= \frac{2750}{139.04}$$= 19.78\text{ mm}$$\Rightarrow t = 20\text{ mm}$
2. Rivet Diameter (d) Unwin's formula for $t = 20\text{ mm}$	$d = 6\sqrt{t}$$= 6\sqrt{20}$$= 26.83\text{ mm}$$\Rightarrow d = 28.5\text{ mm}$
3. Outer Pitch (p) & Efficiency (η) Selected pitch dimensions and verified efficiency	$p_{\text{outer}} = 150\text{ mm}$$p_{\text{inner}} = 75\text{ mm}$$P_t = (p_{\text{outer}} - d)t \cdot \sigma_t$$= (150 - 28.5) \times 20 \times 88$$= 213840\text{ N}$$P = 150 \times 20 \times 88$$= 264000\text{ N}$$\eta = \frac{213840}{264000} \times 100\%$$\Rightarrow \eta = 81\%$
Design Output	Pressure Vessel Joint: Thickness $t = 20\text{ mm}$ Rivet Diameter $d = 28.5\text{ mm}$ Outer Pitch $p = 150\text{ mm}$ Efficiency $\eta = 81\%$

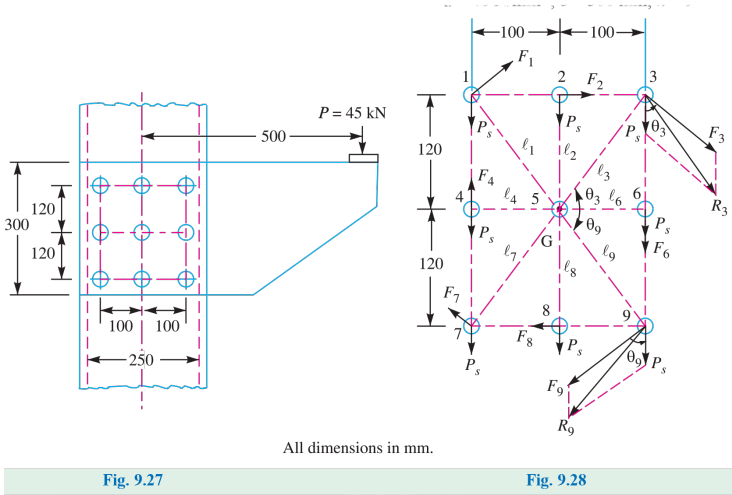


Fig. 9.27 & 9.28: Pressure Vessel Double Strap Butt Joint

SECTION 9: BOILER LONGITUDINAL SEAM DESIGN

System Parameters: Boiler Diameter $D = 1.25\text{ m} = 1250\text{ mm}$, Steam Pressure $p_i = 2.5\text{N/mm}^2$, Ultimate Tensile Stress $\sigma_u = 420\text{ MPa}$, Ultimate Crushing Stress $\sigma_{cu} = 650\text{ MPa}$, Ultimate Shear Stress $\tau_u = 300\text{ MPa}$, Target Efficiency $\eta_l = 80\%$, Factor of Safety $\text{FOS} = 5$, Corrosion Allowance $= 1.5\text{ mm}$.

Design Protocol: Calculate working allowable stresses, determine plate thickness t , and compute rivet diameter d .

1. Allowable Working Stresses Ultimate stress divided by FOS = 5	$\begin{aligned}\sigma_t &= \frac{\sigma_u}{\text{FOS}} \\ &= \frac{420}{5} \\ &= 84\text{ MPa} \\ \sigma_c &= \frac{\sigma_{cu}}{\text{FOS}} \\ &= \frac{650}{5} \\ &= 130\text{ MPa} \\ \tau &= \frac{\tau_u}{\text{FOS}} \\ &= \frac{300}{5} \\ &= 60\text{ MPa}\end{aligned}$
2. Plate Thickness (t) Thin cylinder formula with corrosion allowance	$\begin{aligned}t &= \frac{p_i \cdot D}{2\sigma_t \cdot \eta_l} + 1.5 \\ &= \frac{2.5 \times 1250}{2 \times 84 \times 0.80} + 1.5 \\ &= \frac{3125}{134.4} + 1.5 \\ &= 23.25 + 1.5 \\ &= 24.75\text{ mm} \\ &\Rightarrow t = 25\text{ mm}\end{aligned}$
3. Rivet Diameter (d) Unwin's formula for $t = 25\text{ mm}$	$\begin{aligned}d &= 6\sqrt{t} \\ &= 6\sqrt{25} \\ &= 30\text{ mm} \\ &\Rightarrow d = 30\text{ mm}\end{aligned}$
Design Output	Boiler Seam: Allowable Stresses (84, 130, 60 MPa) Plate Thickness $t = 25\text{ mm}$ Rivet Diameter $d = 30\text{ mm}$

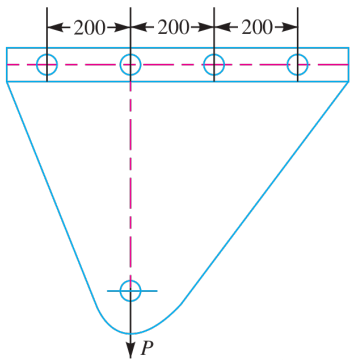


Fig. 9.29

All dimensions in mm.

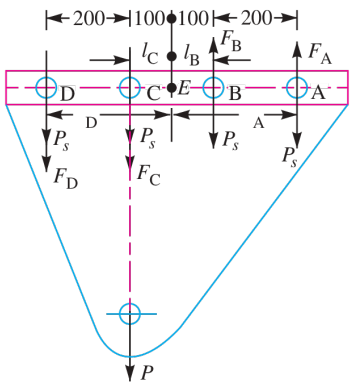


Fig. 9.30

Fig. 9.29 & 9.30: Boiler Longitudinal Seam Triple Riveted Butt Joint

SECTION 10: BOILER LONGITUDINAL & CIRCUMFERENTIAL JOINTS

System Parameters: Boiler Diameter $D = 1.6 \text{ m} = 1600 \text{ mm}$, Pressure $p_i = 2.5 \text{ N/mm}^2$, Allowable Stresses: Tensile $\sigma_t = 75 \text{ MPa}$, Shear $\tau = 60 \text{ MPa}$, Crushing $\sigma_c = 125 \text{ MPa}$.

Design Protocol: Summarize longitudinal joint plate thickness t and rivet diameter d ; design circumferential joint by calculating total steam thrust F and required rivet count N_c .

1. Longitudinal Joint Parameters Plate thickness and rivet diameter selection	$t = 35 \text{ mm}$ $d = 35.5 \text{ mm}$
2. Total Steam End Thrust (F) Axial burst force acting on boiler end cap	$F = \frac{\pi}{4} D^2 \cdot p_i$ $= \frac{\pi}{4} (1600)^2 \times 2.5$ $= 5.026 \times 10^6 \text{ N}$
3. Number of Circumferential Rivets (N_c) Number of single shear rivets required to resist axial thrust	$N_c = \frac{F}{\frac{\pi}{4} d^2 \cdot \tau}$ $= \frac{5.026 \times 10^6}{\frac{\pi}{4} (35.5)^2 \times 60}$ $= \frac{5.026 \times 10^6}{59390}$ $= 84.6$ $\Rightarrow N_c = 85 \text{ rivets}$
Design Output	Longitudinal: $t = 35 \text{ mm}$, $d = 35.5 \text{ mm}$ Circumferential Joint Thrust $F = 5.026 \text{ MN}$ Rivet Count $N_c = 85 \text{ rivets}$

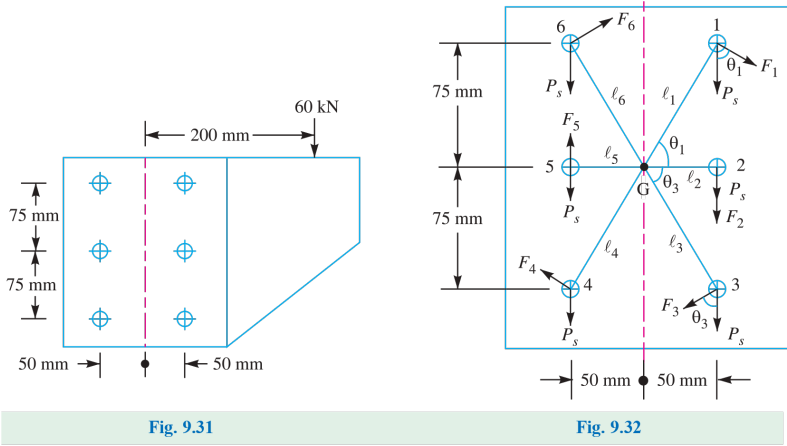


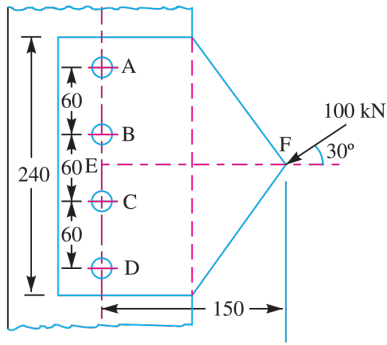
Fig. 9.31 & 9.32: Boiler Longitudinal and Circumferential Joints

SECTION 11: TIE ROD DOUBLE COVER BUTT JOINT

System Parameters: Tie Rod Cross-Section $b = 200\text{ mm}$, $t = 12.5\text{ mm}$, Allowable Stresses: Tensile $\sigma_t = 80\text{ MPa}$, Shear $\tau = 65\text{ MPa}$, Crushing $\sigma_c = 160\text{ MPa}$, Double Shear Factor = 1.875.

Design Protocol: Calculate rivet diameter d , evaluate maximum tensile load capacity P , compute single rivet double shear strength P_s , and determine required rivet count n .

1. Rivet Diameter (d) Unwin’s formula for $t = 12.5\text{ mm}$	$d = 6\sqrt{t}$$= 6\sqrt{12.5}$$= 21.21\text{ mm}$$\Rightarrow d = 21.5\text{ mm}$
2. Full Tensile Load (P) Solid tie-rod tensile capacity	$P = b \cdot t \cdot \sigma_t$$= 200 \times 12.5 \times 80$$= 200000\text{ N}$$= 200\text{ kN}$
3. Single Rivet Double Shear Capacity (P_s) Shear strength per rivet in double cover butt joint	$P_s = 1.875 \cdot \frac{\pi}{4} d^2 \cdot \tau$$= 1.875 \times \frac{\pi}{4} (21.5)^2 \times 65$$= 44250\text{ N}$$= 44.25\text{ kN}$
4. Required Rivet Count (n) Total rivets needed per side of joint	$n = \frac{P}{P_s}$$= \frac{200}{44.25}$$= 4.52$$\Rightarrow n = 5\text{ rivets}$
Design Output	Tie Rod Joint: Rivet Diameter $d = 21.5\text{ mm}$ Tensile Load $P = 200\text{ kN}$ Required Rivets $n = 5\text{ rivets}$



All dimensions in mm.

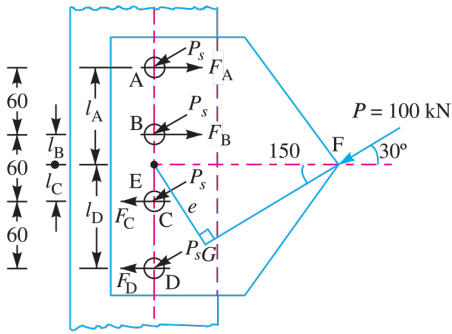


Fig. 9.33

Fig. 9.34

Fig. 9.33 & 9.34: Tie Rod Double Cover Butt Joint

SECTION 12: BRIDGE TIE-BAR LOZENGE (DIAMOND) JOINT

System Parameters: Flat Bar Width $b = 350$ mm, Thickness $t = 20$ mm, Allowable Stresses: Tensile $\sigma_t = 90$ MPa, Shear $\tau = 60$ MPa, Crushing $\sigma_c = 150$ MPa, Double Shear Factor = 1.875.

Design Protocol: Calculate rivet diameter d , evaluate net load capacity P at outer row (1 rivet), determine single rivet double shear strength P_s , compute required number of rivets n , and arrange in diamond pattern.

1. Rivet Diameter (d) Unwin’s formula for $t = 20$ mm	$d = 6\sqrt{t}$$= 6\sqrt{20}$$= 26.83 \text{ mm}$$\Rightarrow d = 27 \text{ mm}$
2. Net Load Capacity (P) Tearing strength at critical Section 1-1 (1 rivet hole)	$P = (b - d)t \cdot \sigma_t$$= (350 - 27) \times 20 \times 90$$= 323 \times 1800$$= 581400 \text{ N}$$= 581.4 \text{ kN}$
3. Single Rivet Double Shear Capacity (P_s) Shear resistance per rivet	$P_s = 1.875 \cdot \frac{\pi}{4} d^2 \cdot \tau$$= 1.875 \times \frac{\pi}{4} (27)^2 \times 60$$= 64412 \text{ N}$$= 64.4 \text{ kN}$
4. Total Rivet Count (n) Total rivets required for lozenge arrangement	$n = \frac{P}{P_s}$$= \frac{581.4}{64.4}$$= 9.02$$\Rightarrow n = 9 \text{ rivets}$
5. Lozenge Rivet Layout Symmetric diamond pattern from outer row to inner row	Arranged in Diamond Pattern: Row 1 (1 rivet), Row 2 (2 rivets), Row 3 (3 rivets), Row 4 (3 rivets)
Design Output	Bridge Lozenge Joint: d = 27 mm Net Capacity P = 581.4 kN Total Rivets n = 9 (Pattern: 1, 2, 3, 3)

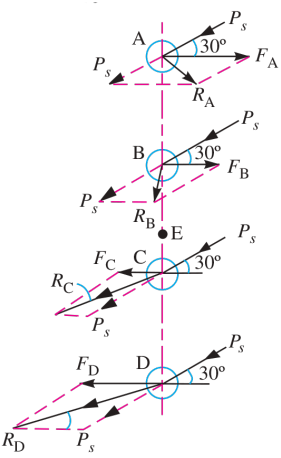


Fig. 9.35

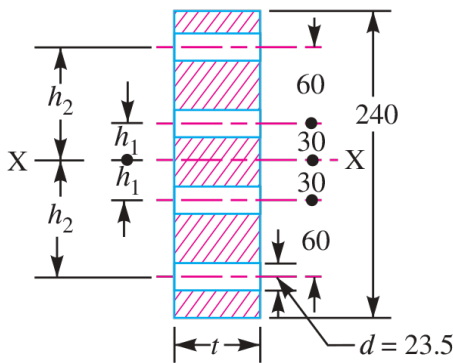
Fig. 9.35: Bridge Tie-Bar Lozenge Joint

SECTION 13: FLAT TIE-BAR LAP JOINT (DIAMOND PATTERN)

System Parameters: Flat Bar Width $b = 200$ mm, Thickness $t = 10$ mm, Rivet Shank Diameter $d = 24$ mm, Rivet Hole Diameter $d_h = 25.5$ mm, Allowable Stresses: Tensile $\sigma_t = 112$ MPa, Shear $\tau = 84$ MPa, Crushing $\sigma_c = 200$ MPa.

Design Protocol: Arrange 4 rivets in diamond pattern (1, 1, 2); evaluate net tearing strength at outer single-rivet row, and compute joint efficiency.

1. Rivet Layout & Pattern Specified rivet arrangement on flat tie-bar	n = 4 rivets arranged in Diamond Pattern (Row 1: 1, Row 2: 1, Row 3: 2)
2. Net Tearing Strength (P_t) Tearing capacity at Section 1-1 with hole diameter $d_h = 25.5 \text{ mm}$	$P_t = (b - d_h)t \cdot \sigma_t$ $= (200 - 25.5) \times 10 \times 112$ $= 174.5 \times 1120$ $= 195440 \text{ N}$ $= 195.44 \text{ kN}$
3. Solid Strength (P) & Efficiency (η) Efficiency based on net tearing capacity	$P = b \cdot t \cdot \sigma_t$ $= 200 \times 10 \times 112$ $= 224000 \text{ N}$ $\eta = \frac{P_t}{P} \times 100\%$ $= \frac{195440}{224000} \times 100\%$ $\Rightarrow \eta = 87.3\%$
Design Output	Flat Lap Joint (Diamond Pattern): Net Tearing Strength Pt = 195.44 kN Joint Efficiency eta = 87.3%



All dimensions in mm.

Fig. 9.36

Fig. 9.36: Flat Tie-Bar Lap Joint (Diamond Pattern)

SECTION 14: ECCENTRIC BRACKET WITH VERTICAL RIVET LINE

System Parameters: Plate Thickness $t = 25$ mm, Eccentric Load $P = 50$ kN = 50000 N, Eccentricity $e = 400$ mm, Rivet Spacing $C = 100$ mm, Number of Rivets $n = 4$ in vertical line, Allowable Shear Stress $\tau = 65$ MPa, Allowable Crushing Stress $\sigma_c = 120$ MPa.

Design Protocol: Calculate direct shear force F_{s1} per rivet; compute secondary shear force F_{s2} due to moment Pe ; find max resultant shear force F_R , and solve for rivet diameter d .

1. Direct Shear Force (F_{s1}) Uniform load distribution across 4 rivets	$F_{s1} = \frac{P}{n}$$= \frac{50000}{4}$$= 12500 \text{ N}$$= 12.5 \text{ kN}$
2. Secondary Shear Force (F_{s2}) Torsional shear force on outermost rivet ($r_{\text{max}} = 150$ mm)	$\sum r_i^2 = 2(150^2 + 50^2)$$= 2(22500 + 2500)$$= 50000\text{mm}^2$$F_{s2} = \frac{P \cdot e \cdot r_{\text{max}}}{\sum r_i^2}$$= \frac{50000 \times 400 \times 150}{50000}$$= 60000 \text{ N}$$= 60 \text{ kN}$
3. Maximum Resultant Shear Force (F_R) Direct plus secondary shear force (co-linear theta = 0 deg)	$F_R = F_{s1} + F_{s2}$$= 12.5 + 60$$= 72.5 \text{ kN}$$= 72500 \text{ N}$
4. Rivet Diameter (d) Equating F_R to rivet single shear strength	$F_R = \frac{\pi}{4}d^2 \cdot \tau$$72500 = \frac{\pi}{4}d^2(65)$$d^2 = \frac{4 \times 72500}{\pi \times 65}$$= \frac{290000}{204.2}$$= 1420.2$$d = 37.68 \text{ mm}$$\Rightarrow d = 38 \text{ mm}$

Design Output	Eccentric Bracket: Direct Shear Fs1 = 12.5 kN Secondary Shear Fs2 = 60 kN Resultant FR = 72.5 kN d = 38 mm
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SECTION 15: ECCENTRICALLY LOADED BRACKET JOINT

System Parameters: Eccentric Load $P = 45\text{ kN}$, Allowable Shear Stress $\tau \leq 40\text{ MPa}$.

Design Protocol: Evaluate direct and secondary shear forces to determine maximum resultant shear load $F_R = 15.2\text{ kN}$; size rivet diameter d .

1. Maximum Resultant Shear Force (F_R) Critical rivet resultant load from direct and secondary shear	$\begin{aligned} F_R &= 15.2\text{ kN} \\ &= 15200\text{ N} \end{aligned}$
2. Rivet Diameter (d) Single shear area calculation for allowable stress $\tau = 40\text{ MPa}$	$\begin{aligned} d &= \sqrt{\frac{4F_R}{\pi\tau}} \\ &= \sqrt{\frac{4 \times 15200}{\pi \times 40}} \\ &= \sqrt{\frac{60800}{125.66}} \\ &= \sqrt{483.8} \\ &= 22.00\text{ mm} \\ &\Rightarrow d = 22\text{ mm} \end{aligned}$
Design Output	Eccentric Bracket (45 kN Load): Critical Resultant FR = 15.2 kN Rivet Diameter d = 22 mm

SECTION 16: MAXIMUM SAFE ECCENTRIC LOAD CAPACITY

System Parameters: Number of Rivets $n = 4$, Rivet Diameter $d = 20\text{ mm}$, Working Shear Stress $\tau = 100\text{ MPa}$.

Design Protocol: Determine maximum single rivet shear load capacity $F_{R,max}$; express critical resultant force as a function of external load P ($F_R = 0.966P$); solve maximum safe load P .

1. Single Rivet Shear Capacity ($F_{R,max}$) Maximum permissible shear force per rivet	$F_{R,max} = \frac{\pi}{4} d^2 \cdot \tau$$= \frac{\pi}{4} (20)^2 \times 100$$= 31416\text{ N}$$= 31.416\text{ kN}$
2. Resultant Load Relation Critical rivet resultant load expressed in terms of load P	$F_R = 0.966P$
3. Maximum Safe Load (P) Equating F_R to single rivet shear capacity	$P = \frac{F_{R,max}}{0.966}$$= \frac{31.416}{0.966}$$= 32.52\text{ kN}$$\Rightarrow P = 32.5\text{ kN}$
Design Output	Eccentric Load Capacity: Single Rivet Capacity $F_{R,max}$ = 31.416 kN Maximum Safe Load P = 32.5 kN

SECTION 17: ECCENTRIC COLUMN BRACKET JOINT

System Parameters: Number of Rivets $n = 6$, Eccentric Load $P = 60 \text{ kN} = 60000 \text{ N}$, Eccentricity $e = 200 \text{ mm}$, Allowable Shear Stress $\tau \leq 150 \text{ MPa}$.

Design Protocol: Calculate primary direct shear F_{s1} and secondary shear F_{s2} ; vectorially combine to get critical resultant shear force F_R ; solve rivet diameter d .

1. Primary & Secondary Shear Forces Direct load per rivet and torsional secondary load	$F_{s1} = \frac{P}{n}$$= \frac{60}{6}$$= 10 \text{ kN}$$F_{s2} = \frac{P \cdot e \cdot r_{\max}}{\sum r_i^2}$$= 41.5 \text{ kN}$
2. Maximum Resultant Shear Force (F_R) Vector addition of primary and secondary shear	$F_R = 51.5 \text{ kN}$$= 51500 \text{ N}$
3. Rivet Diameter (d) Equating F_R to single shear capacity	$d = \sqrt{\frac{4F_R}{\pi \tau}}$$= \sqrt{\frac{4 \times 51500}{\pi \times 150}}$$= \sqrt{\frac{206000}{471.24}}$$= \sqrt{437.1}$$= 20.91 \text{ mm}$$\Rightarrow d = 21 \text{ mm}$
Design Output	6-Rivet Column Bracket: Fs1 = 10 kN Fs2 = 41.5 kN Resultant FR = 51.5 kN Rivet Diameter d = 21 mm

SECTION 18: ECCENTRICALLY LOADED INCLINED BRACKET JOINT

System Parameters: Number of Rivets $n = 4$ (A, B, C, D , vertical line, pitch 60 mm), Inclined Load $P = 100 \text{ kN} = 100000 \text{ N}$ at 30^deg to horizontal, Eccentricity $e = 150 \text{ mm}$, Allowable Shear Stress $\tau = 80 \text{ MPa}$, Allowable Bending Stress $\sigma_b = 125 \text{ MPa}$, Plate Width $b = 240 \text{ mm}$.

Design Protocol: Resolve load into vertical and horizontal components (P_V, P_H); calculate max resultant shear force F_R on critical rivet D to size diameter d ; determine bending moment M on bracket plate to calculate plate thickness t .

1. Vertical & Horizontal Load Components

Resolving inclined 100 kN load at 30 degrees

$$P_V = P \sin(30^\text{deg})$$
$$= 100 \sin(30^\text{deg})$$
$$= 50 \text{ kN}$$
$$P_H = P \cos(30^\text{deg})$$
$$= 100 \cos(30^\text{deg})$$
$$= 86.6 \text{ kN}$$

2. Critical Resultant Shear (F_R) & Rivet Diameter (d)

Resultant force on rivet D and single shear sizing

$$F_R = 42.4 \text{ kN}$$
$$= 42400 \text{ N}$$
$$d = \sqrt{\frac{4F_R}{\pi \tau}}$$
$$= \sqrt{\frac{4 \times 42400}{\pi \times 80}}$$
$$= \sqrt{\frac{169600}{251.33}}$$
$$= \sqrt{674.8}$$
$$= 25.98 \text{ mm}$$
$$\Rightarrow d = \mathbf{26 \text{ mm}}$$

3. Bending Moment (M) & Bracket Thickness (t)

Bending stress analysis of bracket plate

$$M = P_H \cdot e$$
$$= 86.6 \times 10^3 \times 150$$
$$= 1.30 \times 10^7 \text{ N} \cdot \text{mm}$$
$$\sigma_b = \frac{6M}{t \cdot b^2}$$
$$125 = \frac{6 \times 1.30 \times 10^7}{t \times (240)^2}$$
$$= \frac{7.80 \times 10^7}{57600t}$$
$$= \frac{1354.17}{t}$$
$$t = \frac{1354.17}{125}$$
$$= 10.83 \text{ mm}$$
$$\Rightarrow t = \mathbf{14 \text{ mm}}$$

Design Output

Inclined Bracket Joint: PV = 50 kN, PH = 86.6 kN |
Critical FR = 42.4 kN | d = 26 mm | Plate Thickness t =
14 mm